Supplementary Material

When Does Reconfiguration Awareness Pay Off? An Exact-Oracle Study of Learned Link Sleeping in Software-Defined Networks

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These tables complement the results section of the article. All values are generated from the experiment files in `https://github.com/adeliusa486/-lira-sdn-energy` (folder `paper/results`). Section numbers refer to the article.

Table S1: Choice of Model on Nobel-Germany (Base Scenario, Seed 0). Chg/d: Changes per Day

	B4 (snapshot)			LiRA	
Model	Chg/d	J	Latency (ms)	Chg/d	J
XGBoost	12.80	937.2	1.09	0.27	875.9
LightGBM	10.13	924.1	1.61	0.27	875.9
Random forest	0.53	877.2	17.22	0.27	875.9
MLP (2×256)	44.80	1093.7	0.72	0.27	875.9

Table S2: Ablation of LiRA on Nobel-Germany at the Default Price (Mean over Seeds 0 to 2)

Variant	S (%)	Changes/d	MLU viol. (%)	J
Nobel-Germany, high load				
Oracle O*	27.57	0.80	0.00	880.9
LiRA (full)	27.57	0.80	0.00	880.9
no lag features	27.57	0.80	0.00	880.9
no previous config.	27.57	0.80	0.00	880.9
no cost-sensitive weights	27.57	0.80	0.00	880.9
no DAgger	27.57	0.80	0.00	880.9
no safety layer	27.58	0.00	0.14	877.3
Nobel-Germany, stress				
Oracle O*	27.35	0.80	0.00	894.3
LiRA (full)	27.37±0.03	0.93±0.33	0.00	897.4±2.2
no lag features	27.36±0.03	0.84±0.19	0.00	895.6±5.0
no previous config.	27.40	1.07	0.00	902.0
no cost-sensitive weights	27.37±0.03	1.07	0.00	898.6±7.8
no DAgger	27.37±0.03	0.89±0.38	0.00	899.0±9.1
no safety layer	27.40±0.07	0.00	0.17±0.07	889.9±1.6

Table S3: Library Size on Nobel-Germany: Test Coverage and Total Cost J of Each Method (Base Scenario, Seed 0)

K	Test coverage (%)	B4	B6	LiRA	O*
8	100.0	939.3	875.3	875.9	874.7
16	100.0	938.6	875.3	875.9	874.7
32	100.0	937.2	875.3	875.9	874.7
64 (51)	100.0	935.7	875.3	875.8	874.7
128 (51)	100.0	935.7	875.3	875.8	874.7

Table S4: Utilization Threshold θ (Base Scenario, Seed 0)

Topology	θ	All-on viol. (%)	$S_{\mathrm{O}^\star}$ (%)	B4 chg/d	LiRA chg/d	LiRA viol. (%)	LiRA J
Abilene	0.6	0.00	23.98	0.00	0.00	0.00	7226.7
Abilene	0.7	0.00	23.98	0.00	0.00	0.00	7226.7
Abilene	0.8	0.00	23.98	0.00	0.00	0.00	7226.7
Abilene	0.9	0.00	23.98	0.00	0.00	0.00	7226.7
GÉANT	0.6	0.00	29.85	0.00	0.00	0.00	603.7
GÉANT	0.7	0.00	29.85	0.00	0.00	0.00	603.7
GÉANT	0.8	0.00	29.85	0.00	0.00	0.00	603.7
GÉANT	0.9	0.00	29.85	0.00	0.00	0.00	603.7
Nobel-Germany	0.6	0.09	27.63	10.40	0.40	0.00	876.0
Nobel-Germany	0.7	0.05	27.63	12.80	0.27	0.00	875.9
Nobel-Germany	0.8	0.05	27.63	12.80	0.27	0.00	875.9
Nobel-Germany	0.9	0.05	27.63	12.80	0.27	0.00	875.9

Table S5: Per-Snapshot Optimization Baselines on a Test Subsample

Topology	Method	n	S (%)	Median time (s)
Abilene	Library snapshot optimum	60	23.98	–
Abilene	B1 per-snapshot ILP	60	23.98	0.53
Abilene	B3 NSGA-II	15	23.98	253.97
Abilene	B5 greedy	60	23.55	0.04
GÉANT	Library snapshot optimum	60	29.85	–
GÉANT	B1 per-snapshot ILP	60	29.85	3.53
GÉANT	B3 NSGA-II	60	29.56	41.56
GÉANT	B5 greedy	60	27.16	0.24
Nobel-Germany	Library snapshot optimum	60	27.63	–
Nobel-Germany	B1 per-snapshot ILP	60	27.63	0.54
Nobel-Germany	B3 NSGA-II	60	27.63	28.34
Nobel-Germany	B5 greedy	60	27.29	0.09

Table S6: Scalability of the Optimization Step and the Safety Check

Topology	$\|V\|$	$\|E\|$	OD pairs	ILP (s)	Greedy (s)	Safety check (ms)
Abilene	12	15	132	0.48	0.04	0.067
Nobel-Germany	17	26	272	0.41	0.19	0.130
GÉANT	22	36	462	4.07	0.56	0.157
Germany50	50	88	2450	1.61	1.45	0.223
TA2	65	108	4160	11.37	7.42	0.363

Table S7: Repetition of the Main Study Without a Wear Price ($\lambda_W = 0$, All Methods Retrained): Configuration Changes per Day and Cost J (Energy plus Reconfiguration). B4, B6, and LiRA: Mean over Five Seeds

		Changes/d				Cost J with $\lambda_W = 0$					
Topology	Scenario	B4	B6	LiRA	O*	B4	B6	B2	Static	LiRA	O*
Abilene	base	0.02	0.00	0.00	0.00	9151.3	9151.0	9154.7	9151.0	9151.0	9151.0
Abilene	high load	0.00	0.00	0.00	0.00	9157.3	9157.3	9161.0	9157.3	9157.3	9157.3
Abilene	high dynamic power	0.02	0.00	0.00	0.00	9172.3	9172.0	9183.1	9172.0	9172.0	9172.0
Abilene	stress	0.00	0.00	0.00	0.00	9190.9	9190.9	9201.5	9190.9	9190.9	9190.9
GÉANT	base	0.00	0.00	0.00	0.00	1897.8	1897.8	1897.8	1897.8	1897.8	1897.8
GÉANT	high load	0.00	0.00	0.00	0.00	1904.4	1904.4	1904.4	1904.4	1904.4	1904.4
GÉANT	high dynamic power	0.00	0.00	0.00	0.00	1919.1	1919.1	1919.1	1919.1	1919.1	1919.1
GÉANT	stress	0.00	0.00	0.00	0.00	1938.9	1938.9	1938.9	1938.9	1938.9	1938.9
Nobel-Germany	base	11.68	0.13	0.27	0.00	1582.5	1566.0	1566.0	1566.1	1566.1	1565.9
Nobel-Germany	high load	1.76	1.07	0.88	0.80	1572.4	1571.1	1570.0	1569.6	1570.2	1569.6
Nobel-Germany	high dynamic power	11.68	0.27	0.27	0.27	1590.3	1573.9	1574.0	1573.9	1573.9	1573.9
Nobel-Germany	stress	1.76	1.07	0.93	0.80	1584.8	1583.5	1583.1	1584.1	1583.7	1583.0

Table S8: Injected Traffic Shifts: Post-Shift Cost J of LiRA Trained Once ("Once") and with Drift-Triggered Retraining ("Retrained"). FA/1k Is the Number of False Alarms per 1000 Intervals on the Stream Without a Shift

Topology	Shift	J once	J retrained	Detections	FA/1k
Abilene	growth	4701.5	4701.5	2	0.17
Abilene	hotspot	4698.9	4698.9	2	0.17
Abilene	phase	4698.2	4698.2	2	0.17
GÉANT	growth	394.8	394.8	1	0.00
GÉANT	hotspot	392.6	392.6	0	0.00
GÉANT	phase	392.3	392.3	0	0.00
Nobel-Germany	growth	569.9	570.6	2	0.93
Nobel-Germany	hotspot	573.9	574.5	2	0.93
Nobel-Germany	phase	569.6	570.3	2	0.93

Table S9: Rolling-Origin Folds (Mean over Four Folds, Default Price)

Topology	Method	S (%)	Chg/d	J (mean±std over folds)
Abilene	B4	23.98	0.00	2890.5±0.9
	B6	23.98	0.00	2890.5±0.9
	LiRA	23.98	0.00	2890.5±0.9
	O*	23.98	0.00	2890.5±0.9
GÉANT	B4	29.84	0.00	241.7±0.7
	B6	29.84	0.00	241.7±0.7
	LiRA	29.84	0.00	241.7±0.7
	O*	29.84	0.00	241.7±0.7
Nobel-Germany	B4	27.64	11.17	371.5±5.4
	B6	27.64	0.25	350.1±0.5
	LiRA	27.64	0.33	350.3±0.7
	O*	27.63	0.17	350.0±0.5

Table S10: Mininet Emulation of the Most Active Day of Nobel-Germany (B4: Three Runs; B0 and LiRA: One Run). Trans.: Configuration Changes. Loss in Windows Counts Probes Sent from 0.2 s Before a Change Starts to 1 s After It Ends. Link-Hours Are Computed from the Replayed Configurations

Method	Runs	Trans.	Rule install (ms)	Trans. time (s)	Loss (%)	Loss in windows (%)	Delay p50 (ms)	p99 (ms)	Link-hours
B0	1	0	–	–	0.000	–	0.10	0.4	4.16
B4	3	18	99	1.11	0.014	0.22	0.10	0.3	2.88
LiRA	1	0	–	–	0.000	–	0.09	0.2	2.88